Beaconhouse National University

BNU

Assignment: 01

Semester: 01

Department: SCIT (BSCS)

Course: Applied Physics

Submitted by: Saad Mughal

Submitted to: Sir Khavar Zia

Student ID: F2023-009

Section: B

Submission Date: 24-OCT-2023

Video 1: https://www.youtube.com/watch?v=ErcH_OuCaNY

Topic: Intrinsic and Extrinsic Semiconductors
Playing Time: 5:11 min
Description: This video teaches Intrinsic and Extrinsic semiconductors and their difference. When heat is applied to Silicon, the covalent bonds break and atoms jump from valence band to conduction band leaving a hole in the valence band. This is the reason that semiconductors conduct electricity but it is very small so another substance is added in silicon to make it Extrinsic from Intrinsic which is called doping. Doping with Trivalent elements like Aluminium gives P-Type semiconductor which has majority carriers as holes and when Silicon is doped with Pentavalent element then the semiconductor becomes N-Type semiconductor that has majority carriers as electrons. Extrinsic semiconductor conducts more electricity than Intrinsic semiconductor.

Video 2: https://www.youtube.com/watch?v=YsrAoClhROc

Topic: Drift and Diffusion
Playing Time: 10:48 min
Description: This video teaches how Drift and Diffusion works. When P & N-Type are joined then make a P-N Junction. Holes from P-type are diffused to N-type and electrons from N-type are diffused to P-type. If we look closely, we can see that the P-N junction is in 3 pieces, P-type, N-type and in the middle there's Depletion region in which there's an electric field with direction N to P. The equilibrium is always maintained between N and P.
Diffusion Current occurs from P to N due to majority charge carriers. Drift current occurs from N to P due to minority carriers. Diffusion and Drift Current are always in equilibrium. There are no charge carriers in depletion region.

Video 3: https://www.youtube.com/watch?v=Bu52CE55BN0

Topic: Working of Semiconductor Fab
Playing Time: 7:43 min
Description: This video teaches how semiconductor is manufactured. The main material used to make semiconductors is silicon. It needs to go through various processes that are:
Wafer Manufacturing: Wafers are the foundation of a semiconductor, wafers are made from silicon extraction from sand.
Oxidation: For wafer to be conductive it needs to go through the oxidation process where water vapor and oxygen are sprayed on the wafer disk, this layer protects the surface of the wafer and blocks leakage current between circuits.
Photolithography: The process to draw circuit design on a wafer. Photomask is a glass substrate with a computer designed circuit pattern.
Etching: The process to carve out unnecessary materials so that only the design pattern remains.
Deposition & Ion implantation
Metal wiring

Electrical Die Sorting
Packaging

I couldn’t understand the process further. That's why I only wrote till Etching.

Video 4: https://www.youtube.com/watch?v=IYV_nVY2caQ

Topic: Rutherford and Bohr Models of the Atom
Playing Time: 6:16 min
Description: This video teaches us the comparison between Bohr and Rutherford Models.
The Rutherford model states that the nucleus is packed with neutrons and protons, and overall positively charged. The nucleus is surrounded by a large and diffuse electron cloud. The total charges of the electron exactly balance the positive charge of the nucleus. Though it fails to explain the arrangement of electrons.
The Bohr model states that electrons are found in the orbits. He introduced the idea of quantized energy levels in the atom and provided a reasonable explanation for the discrete atomic line spectra in terms of electron energies. Although he failed to explain the arrangement of electrons and chemical properties of atoms.

Video 5: https://www.youtube.com/watch?v=y865nLVyQsY

Topic: Conductors, Insulators and Semiconductors
Playing Time: 8:02 min
Description: This video teaches the difference between conductors, insulators and semiconductors using an energy level diagram.
Conductors: Valence and conduction bands overlap so we can say that there is only one band in conductors such is the reason for high conductivity.
Insulateros: The distance between valence and conduction band is massive (>5eV) therefore it can not conduct any electricity.
Semiconductors: There is distance between valence band and conduction band but it’s (<3eV) therefore if heat is applied, electrons can jump from valence band to conduction band leaving holes in valence band and in conclusion it becomes a semiconductor.